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The Harrow-Hassidim-Lloyd algorithm

PhD course "The ABCs of Quantum Computing"

Marco Galliani

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Harrow et al.2009

Let $Ax = b$ a linear system, $A \in \mathbb{R}^{N_b \times N_b}$, $\mathbf{b} \in \mathbb{R}^{N_b}$

- **Input limitations:** A sparse and hermitian
- **Output limitations:**
 - ▶ HHL **does not** actually compute x (*amplitude encoding*)
 - ▶ HHL computes a *feature* of x , e.g., $x^\top Mx$
- **Logarithmic scaling in N_b :**

$$\mathcal{O}(\log N_b \kappa^2 / \epsilon) \quad \text{vs} \quad \mathcal{O}(N_b \kappa)$$

κ : condition number of A , ϵ : desired accuracy

Given the eigenvalue decomposition (EVD) of the matrix, $A = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$, its inverse can be computed as

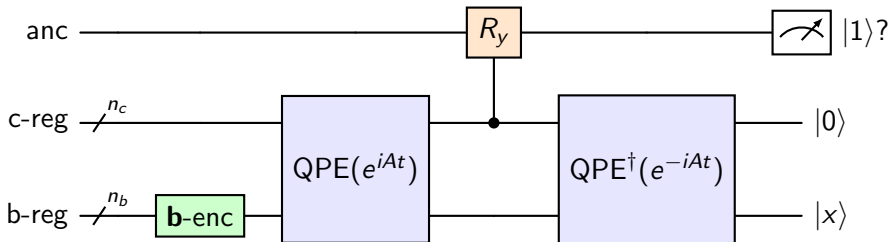
$$A^{-1} = \sum_i \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^\top$$

Thus, if we rewrite $\mathbf{b}$ in the eigenvector basis, $\mathbf{b} = \sum_i \beta_i \mathbf{u}_i$, $\mathbf{x}$ will be

$$\mathbf{x} = A^{-1} \mathbf{b} = \sum_i \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^\top \mathbf{b} = \sum_i \lambda_i^{-1} \beta_i \mathbf{u}_i$$

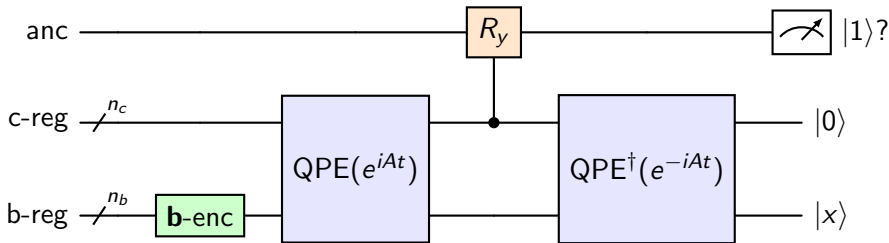
Registers:

- *b*-register: components of $\mathbf{b}, \mathbf{x}$, (*amplitude encoding*), n_b -qubits, with $N_b = 2^{n_b}$
- *c*-register: phase of the eigenvalues of A
- *ancilla*: inversion and measurement



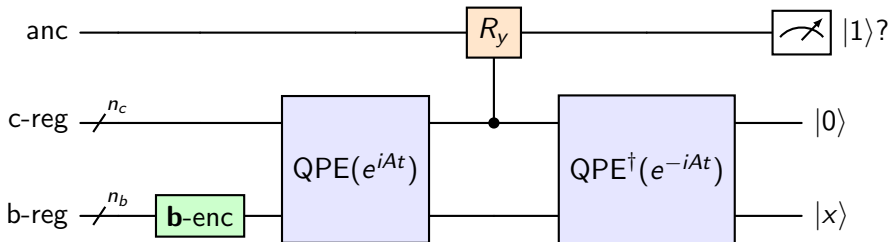
Initial state:

$$|\psi_0\rangle = |0\rangle^{\otimes n_b} |0\rangle^{\otimes n_c} |0\rangle$$



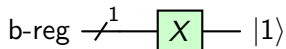
Steps:

1. State preparation
2. QPE
3. Ancilla qubit rotation
4. Inverse QPE



State preparation: rotate the qubits in the b-register to have amplitudes corresponding to the coefficients of **b** (*amplitude encoding*)

An example: $\mathbf{b} = (0, 1)$, $N_b = 2 \implies n_b = 1$



Recall:

- state preparation step specific to each **b**
- **b** needs to be normalized: $\sum_i |\tilde{b}_i|^2 = 1$
- output: $|\Psi_0\rangle = |\mathbf{b}\rangle |0\rangle^{\otimes n_c} |0\rangle$

Goal: estimate the phase of the eigenvalues of A

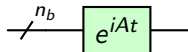
Issues:

- How to encode matrix A in the circuit? *Hamiltonian encoding*
- How to estimate its eigenvalues? *Quantum Phase Estimation*

In general, A non-unitary: we'll use

$$U = e^{iAt}$$

which is unitary and can be approximated by solving a *Hamiltonian simulation* problem



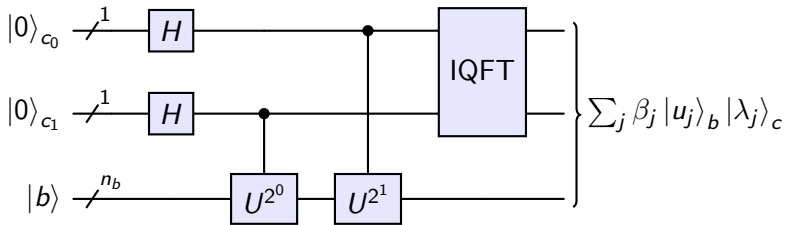
Problem (Hamiltonian Simulation Problem)

Given a Hamiltonian ($2^{n_b} \times 2^{n_b}$ Hermitian matrix), A , a time t and a tolerance ϵ , find U such that

$$\|U - e^{iAt}\| \leq \epsilon$$

Algorithm: Trotterization + Pauli gadget ($A = \sum_j A_j$, $U \approx \prod_j e^{iA_j t}$)

Eigenvalue connection: $\text{eig}(U) = e^{i\lambda_j t}$, where λ_j eigenvalues of A



Steps:

1. Superposition
2. Controlled rotations
3. Inverse Quantum Fourier Transforms (IQFT)

After the superposition:

$$|\psi_2\rangle = |b\rangle \frac{1}{2^{\frac{n_C}{2}}} (|0\rangle + |1\rangle)^{\otimes n} |0\rangle$$

For the moment, let's assume $|b\rangle$ eigenvector of U with eigenvalue $e^{2\pi i\phi}$ (we can always express $|b\rangle = \sum_j \beta_j |u_j\rangle$), then:

$$U |b\rangle = e^{2\pi i\phi} |b\rangle$$

If the control is 0, $|b\rangle$ is not affected, otherwise U is applied. This is equivalent to (*phase kickback*):

$$\begin{aligned}
 |\psi_3\rangle &= |b\rangle \otimes \left(\frac{1}{2^{\frac{n_c}{2}}} (|0\rangle + e^{2\pi i \phi 2^{n_c-1}} |1\rangle) \otimes \cdots \otimes (|0\rangle + e^{2\pi i \phi 2^0} |1\rangle) \right) \otimes |0\rangle_a \\
 &= |b\rangle \otimes \left(\frac{1}{2^{\frac{n_c}{2}}} \bigotimes_{j=1}^{n_c} (|0\rangle + e^{2\pi i \phi 2^{n_c-j}} |1\rangle) \right) \otimes |0\rangle_a \\
 &= |b\rangle \otimes \left(\frac{1}{2^{\frac{n_c}{2}}} \sum_{k=1}^{2^{n_c}-1} e^{2\pi i \phi k} |k\rangle \right) \otimes |0\rangle_a
 \end{aligned}$$

Motivation: transfer phase information from U to the control-qubits (*c-registers*)!

To extract the phase information from the c -registers of $|\Psi_3\rangle$, IQFT is applied:

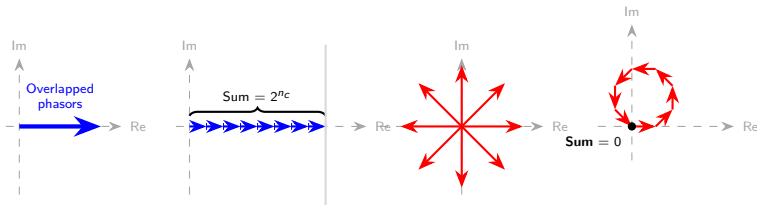
$$\begin{aligned}
 \text{IQFT}\left(\frac{1}{2^{\frac{n_c}{2}}} \sum_{k=0}^{2^{n_c}-1} (e^{2\pi i \phi k} |k\rangle)\right) &= \frac{1}{2^{\frac{n_c}{2}}} \sum_{k=0}^{2^{n_c}-1} (e^{2\pi i \phi k} \text{IQFT}(|k\rangle)) \\
 &= \frac{1}{2^{\frac{n_c}{2}}} \sum_{k=0}^{2^{n_c}-1} (e^{2\pi i \phi k} \left(\frac{1}{\sqrt{2^{n_c}}} \sum_{j=0}^{N_c-1} e^{-2\pi i j k / N_c} |j\rangle\right)) \\
 &= \frac{1}{2^{n_c}} \sum_{k=0}^{2^{n_c}-1} \left(\sum_{j=0}^{2^{n_c}-1} e^{2\pi i k(\phi - j/N_c)} |j\rangle\right)
 \end{aligned}$$

Given the state before measurement:

$$\frac{1}{2^{n_c}} \sum_{k=0}^{2^{n_c}-1} \left(\sum_{j=0}^{2^{n_c}-1} e^{2\pi i k(\phi - j/N_c)} |j\rangle \right)$$

The following alternatives apply:

- **Constructive** (if $\phi - j/N_c = 0$): Amplitude proportional to $\sum e^0 = 2^{n_c}$
- **Destructive** (otherwise): Amplitude cancels since $\sum e^{2\pi i k(\dots)} = 0$



Finally, ignoring the states with zero-amplitude:

$$\begin{aligned} |\Psi_4\rangle &= \frac{1}{2^n} |b\rangle \sum_{k=0}^{2^n-1} e^{2\pi i k \cdot 0} |N_c \phi\rangle |0\rangle_a \\ &= |b\rangle |N_c \phi\rangle |0\rangle_a \end{aligned}$$

The *c-registers* qubits are used to represent the phase information of U !

Recap:

- We assumed: $|b\rangle$ eigenvector of U with eigenvalue $e^{2\pi i\phi}$
- U encodes A as $U = e^{iAt}$, so that $U|b\rangle = e^{i\lambda_j t}|b\rangle$

Thus, $\phi = \frac{\lambda_j t}{2\pi}$ and

$$|\Psi_4\rangle = |u_j\rangle \left| \frac{N_c \lambda_j t}{2\pi} \right\rangle |0\rangle_a$$

More generally, writing $|b\rangle = \sum_j \beta_j |u_j\rangle$

$$|\Psi_4\rangle = \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle \left| \frac{N_c \lambda_j t}{2\pi} \right\rangle |0\rangle_a$$

Note: $\left| \frac{N_c \lambda_j t}{2\pi} \right\rangle$ encoded through basis-encoding. t, N_c to make the latter integers

Rotation of the ancilla qubit: eigenvalue inversion, depends on a parameter C

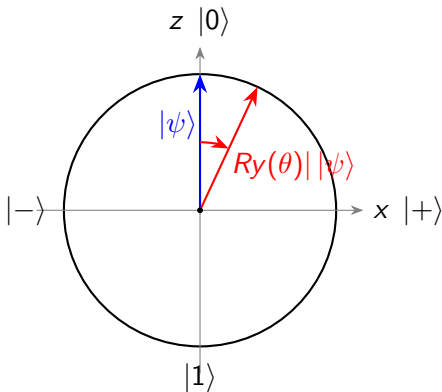
$$|\psi_5\rangle = \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle_a + \frac{C}{\tilde{\lambda}_j} |1\rangle_a \right)$$

Ry-gates:

$$R_y(\theta_j) = \begin{bmatrix} \cos(\theta_j/2) & -\sin(\theta_j/2) \\ \sin(\theta_j/2) & \cos(\theta_j/2) \end{bmatrix}$$

and angle

$$\theta_j = \arcsin\left(\frac{C}{\lambda_j}\right)$$



Once the ancilla qubit is measured we got two alternatives

- if $|0\rangle$: throw away the results
- if $|1\rangle$: store the results

After that the state is updated to

$$|\psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N_c}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} b_j |u_j\rangle |\tilde{\lambda}_j\rangle \frac{C}{\tilde{\lambda}_j} |1\rangle_a$$

$$|\psi_6\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N_c}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle |\tilde{\lambda}_j\rangle |1\rangle_a$$

Issues:

- b-registers are not encoded with respect to the canonical basis
- b-registers and c-registers are entangled

To do so, QFT is applied to the *c-register* qubits:

$$\text{QFT}(|\tilde{\lambda}_j\rangle) = \frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{2\pi i y \tilde{\lambda}_j / N_c} |y\rangle$$

leading to

$$|\Psi_7\rangle = \frac{1}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N_c}{\tilde{\lambda}_j} \right|^2} \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{2\pi i y \tilde{\lambda}_j / N_c} |y\rangle \right) |1\rangle_a$$

Then controlled rotations with U^{-1} are applied as before

$$\begin{aligned} & \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \left(\frac{1}{2^{\frac{n}{2}}} \sum_{y=0}^{2^n-1} e^{-i\lambda_j t y} e^{2\pi i y \tilde{\lambda}_j / N_c} |y\rangle \right) \\ &= \sum_{j=0}^{2^{n_b}-1} C \frac{b_j}{\tilde{\lambda}_j} |u_j\rangle \sum_{y=0}^{2^n-1} |y\rangle = C |x\rangle \sum_{y=0}^{2^n-1} \frac{1}{2^{\frac{n}{2}}} |y\rangle \end{aligned}$$

leading to

$$|\Psi_8\rangle = \frac{C}{2^{\frac{n}{2}} \sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N_c}{\tilde{\lambda}_j} \right|^2} |x\rangle \sum_{y=0}^{2^n-1} \frac{1}{2^{\frac{n}{2}}} |y\rangle |1\rangle_a$$

The clock-qubits and *b-registers* are now unentangled. By applying the Hadamard gates to the clock qubits we have

$$|\psi_9\rangle = \frac{C}{\sum_{j=0}^{2^{n_b}-1} \left| \frac{b_j N_c}{\tilde{\lambda}_j} \right|^2} |x\rangle_b |0\rangle_c^{\otimes n} |1\rangle_a$$

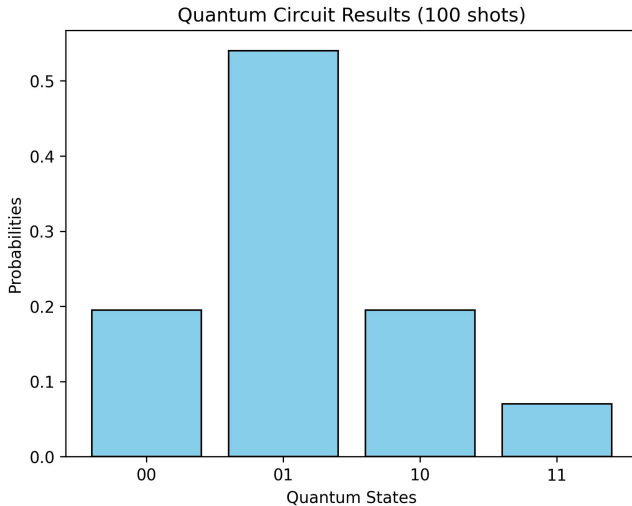
where the solution $|x\rangle$ is stored in the *b-registers* successfully.

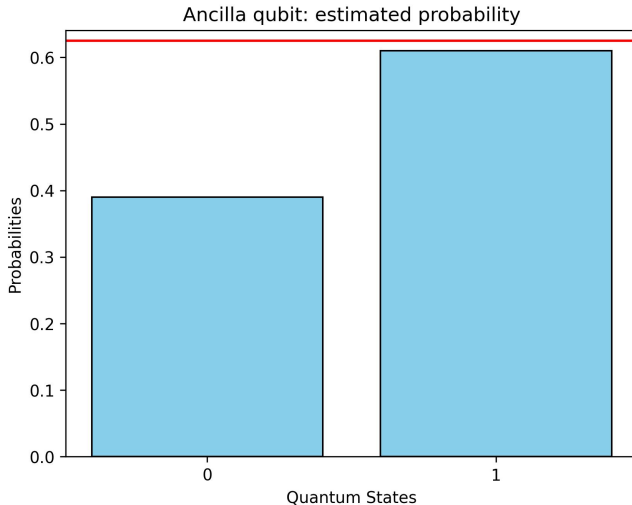
Linear system, $Ax = b$

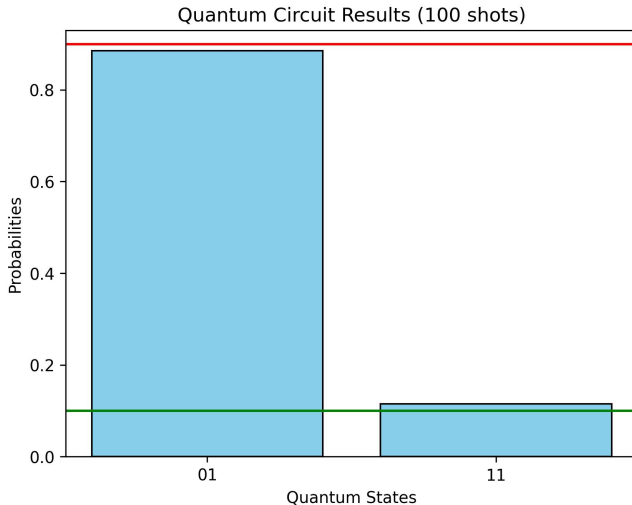
$$A = \begin{bmatrix} 1 & -1/3 \\ -1/3 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

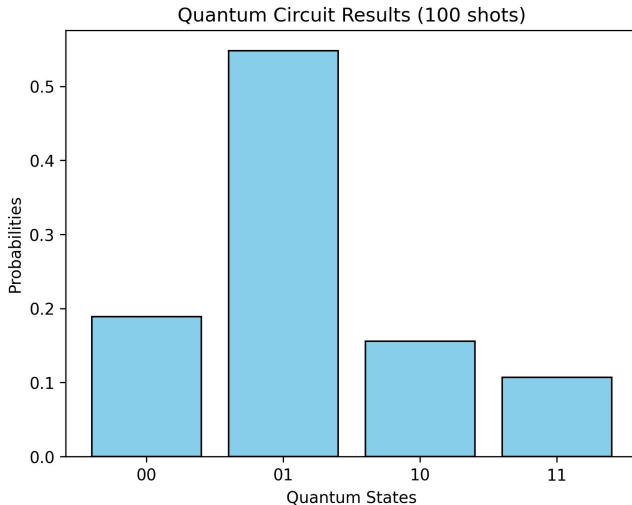
Actual solution

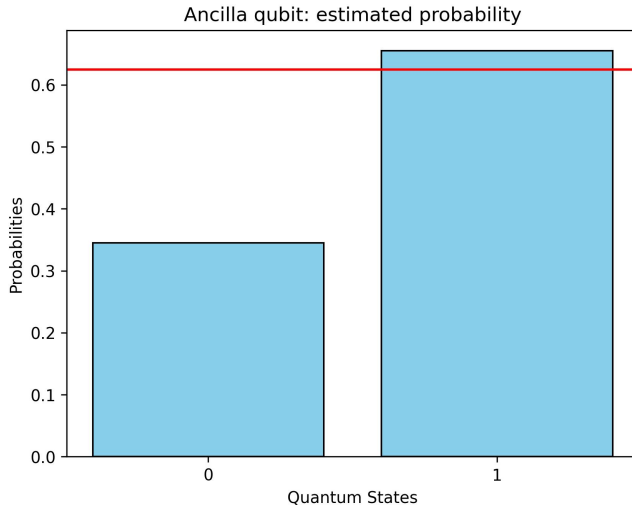
$$x = \begin{bmatrix} 1.125 \\ 0.375 \end{bmatrix}$$

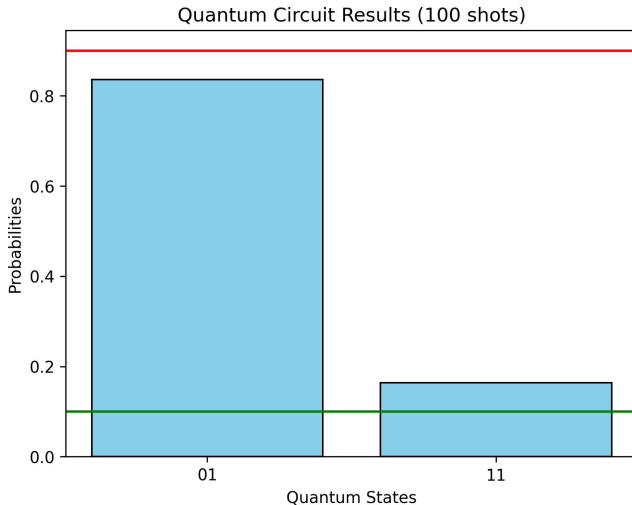












A. W. Harrow, A. Hassidim, and S. Lloyd. Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15), Oct. 2009. ISSN 1079-7114. doi: 10.1103/physrevlett.103.150502. URL <http://dx.doi.org/10.1103/PhysRevLett.103.150502>.